
Quantum Computation and Communications (EE798V):

Problem Set 4

Department of Electrical Engineering, IIT Kanpur

Q. 1: All quantum algorithms that we studied in the class presuppose the existence of a quantum oracle U_f that can simulate the behavior of a classical Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ in the following manner:

$$U_f |x\rangle |b\rangle = |x\rangle |b \oplus f(x)\rangle, \text{ where } x \in \{0, 1\}^n, b \in \{0, 1\}$$

- (a) Argue that U_f can be represented by a $2^{n+1} \times 2^{n+1}$ matrix whose each column is a one-hot vector.
- (b) Show that U_f is an injective (one-to-one) map i.e., $U|x_1\rangle|b_1\rangle = U|x_2\rangle|b_2\rangle$ if and only if $x_1 = x_2$ and $b_1 = b_2$ where $x_1, x_2 \in \{0, 1\}^n$ and $b_1, b_2 \in \{0, 1\}$.
- (c) Combine the above results to conclude that U_f is a unitary matrix.
- (d) Construct an explicit quantum circuit to simulate the behavior of the AND function $f(x_1, x_2) = x_1 x_2$ where $x_1, x_2 \in \{0, 1\}$. (Hint: Use the Toffoli gate).
- (e) Construct an explicit quantum circuit to simulate the behavior of the OR function $f(x_1, x_2) = x_1 + x_2$ where $+$ indicates the OR operation and $x_1, x_2 \in \{0, 1\}$. (Hint: Use a CNOT gate to simulate a NOT function and apply the De Morgan's identity.)

Q. 2: Recall the transformation D_N used in Grover's algorithm where $N = 2^n$ for some integer n , and

$$D_N = -I + 2|\phi_0\rangle\langle\phi_0|, \text{ where } |\phi_0\rangle = \frac{1}{\sqrt{N}} \sum_{x \in \{0, 1\}^n} |x\rangle$$

- (a) Show that $\langle\phi_0|\phi_0\rangle = 1$.
- (b) Using (a), prove that D_N is unitary.
- (c) Show that $\langle\phi_0|y\rangle = 1/\sqrt{N}$ for all $y \in \{0, 1\}^n$.

Let S be a subset of $\{0, 1\}^n$ with cardinality $S < 2^n$. Define the following vectors.

$$|x_S\rangle = \frac{1}{\sqrt{S}} \sum_{x \in S} |x\rangle, \quad |x_S^\perp\rangle = \frac{1}{\sqrt{N-S}} \sum_{x \notin S} |x\rangle$$

- (d) Show that $|x_S\rangle$ and $|x_S^\perp\rangle$ are orthonormal vectors.
- (e) Express $|\phi_0\rangle$ as a linear combination of $|x_S\rangle$ and $|x_S^\perp\rangle$.
- (f) Using (d) and (e), compute the inner products $\langle\phi_0|x_S\rangle$ and $\langle\phi_0|x_S^\perp\rangle$ in terms of S and N .
- (g) Using (e) and (f), express $D_N|x_S\rangle$ and $D_N|x_S^\perp\rangle$ as linear combinations of $|x_S\rangle$ and $|x_S^\perp\rangle$.
- (h) Consider an arbitrary state $|\psi\rangle = \alpha|x_S\rangle + \beta|x_S^\perp\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$. Using (g), express $D_N|\psi\rangle$ as a linear combination of $|x_S\rangle$ and $|x_S^\perp\rangle$.

Q. 3: An n -qubit Inverse Quantum Fourier Transform (IQFT) is defined as follows.

$$|j\rangle \rightarrow \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \exp\left(-\frac{2\pi i j k}{N}\right) |k\rangle, \text{ where } j, k \in \{0, \dots, N-1\}, N = 2^n$$

- (a) Let $\omega = \exp(-2\pi i/N)$. Write the matrix describing the above transformation in terms of ω .
- (b) Prove that the above matrix is unitary.
- (c) Show that the above transformation can be equivalently written as follows.

$$|j\rangle \rightarrow \frac{1}{\sqrt{2}} \left(|0\rangle + \exp(-2\pi i 0.j_n) |1\rangle \right) \otimes \frac{1}{\sqrt{2}} \left(|0\rangle + \exp(-2\pi i 0.j_{n-1}j_n) |1\rangle \right) \otimes \dots \otimes \frac{1}{\sqrt{2}} \left(|0\rangle + \exp(-2\pi i 0.j_1 \dots j_n) |1\rangle \right)$$

where $j_1 \dots j_n$ is the binary representation of j .

- (d) Using the above result, construct a quantum circuit for computing 3-qubit IQFT using only 1-qubit and 2-qubit gates. Properly define and label all the gates used.
- (e) Taking hints from the above construct, write the total number of 1-qubit and 2-qubit gates needed to perform an n -qubit IQFT.
- (f) Explicitly compute the 3-qubit IQFT of $|010\rangle$ using the result (c).

Q. 4: Consider two integers x, N such that $(x, N) = 1$ i.e., they do not share any common factors other than 1. Let $L = \lceil \log_2 N \rceil$ i.e., $2^{L-1} < N \leq 2^L$. Define the following function.

$$U|y\rangle = \begin{cases} |xy(\bmod N)\rangle & \text{if } 0 \leq y \leq N-1 \\ |y\rangle & \text{if } N \leq y \leq 2^L-1 \end{cases} \quad (1)$$

- (a) Show that U can be represented by a $2^L \times 2^L$ matrix whose each column is a one-hot vector.
- (b) Show that U is injective i.e., $U|y_1\rangle = U|y_2\rangle$ if and only if $y_1 = y_2$ where $y_1, y_2 \in \{0, \dots, 2^L-1\}$.
- (c) Combine the above results to conclude that U is unitary.

Let r be the order of x modulo N i.e., r is the smallest positive integer such that $x^r = 1(\bmod N)$. Consider the following state:

$$|\phi_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} \exp\left(-\frac{2\pi i s k}{r}\right) |x^k(\bmod N)\rangle, \quad 0 \leq s \leq r-1$$

- (d) Show that $x^p(\bmod N) \equiv x^q(\bmod N)$ if and only if $p = q$ where $p, q \in \{0, \dots, r-1\}$.
- (e) Use the above result to show that $\langle \phi_s | \phi_s \rangle = 1$.
- (f) Show that $|\phi_s\rangle$ is an eigenvector of U with $\exp(2\pi i s/r)$ as its corresponding eigenvalue.
- (g) Let $s/r = 0.m_1m_2 \dots m_t$. Show that $|\phi_s\rangle$ is also an eigenvector of U^{2^j} with $\exp(2\pi i 0.m_{j+1} \dots m_t)$ as its associated eigenvalue.
- (h) Explicitly describe U^{2^j} in the form of (1).
- (i) Assuming access to $|\phi_s\rangle$, construct a quantum circuit to estimate $s/r = 0.m_1m_2 \dots m_t$. Since s is known, we can compute r from this circuit.
- (j) Prove the following equality for any $k \in \{0, 1, \dots, r-1\}$.

$$\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} \exp\left(\frac{2\pi i s k}{r}\right) |\phi_s\rangle = |x^k(\bmod N)\rangle$$

- (k) As a special case of the above result, show that

$$\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |\phi_s\rangle = |1\rangle$$

It is to be noted that here, $|1\rangle$ is an element of a Hilbert space of dimension 2^L . Express $|1\rangle$ in the vector form.

- (l) Assume, for simplicity, that s/r is expressible in t -bits for any $s \in \{0, \dots, r-1\}$. Apply the above result to modify the circuit in (i) to compute the order r without assuming access to any eigenstate.

Q. 5: Assume that we wish to factor $N = 91$ utilizing Shor's algorithm. Suppose that we choose $x = 4$ as a number in $\{2, \dots, 90\}$.

- (a) Use Euclid's algorithm to show that $\gcd(4, 91) = 1$.
- (b) Find the order, r , of x modulo N .
- (c) Show that r is even and $x^{r/2} \not\equiv -1(\bmod N)$.
- (d) Use Euclid's algorithm to compute $\gcd(x^{r/2} + 1, N)$.
- (e) Confirm that the result computed in (d) is a factor of N .